Science — Unit 4, electricity and thunderstorms · Grade 6

Storm Lab: The Weather Station

Answer the 12 questions. Use the observations given; you do not need to perform an experiment. Allow 10 to 15 minutes.

> You are reviewing a weather station's training cards. Practise conductors, insulators, experiments, and thunderstorm safety.

1 What is the name of the tall storm clouds discussed in Unit 4? Write the cloud name only.

> **Answer:** cumulonimbus
> Cumulonimbus clouds are thunderstorm clouds.

2 For flash A, you hear thunder 3 seconds later. For flash B, you hear thunder 9 seconds later. Which flash was farther away? Write A or B.

> **Answer:** B
> A longer delay means that the sound travelled farther. Flash B was farther away. Both storms still require shelter.

3 Why do you usually see a flash before hearing its thunder?

- Light travels faster than sound.
- Sound travels faster than light.
- The flash waits for the thunder.

> **Answer:** Light travels faster than sound.
> The light reaches you sooner. The sound takes longer to travel from the discharge to you.

4 What helps separate electrical charges inside a storm cloud?

- The cloud touching a metal roof
- Collisions between small ice crystals and soft hail
- Two whole clouds rubbing together like balloons

> **Answer:** Collisions between small ice crystals and soft hail
> Particles collide inside the cloud. These collisions, together with air movements, help separate charges. Whole clouds do not need to rub against each other.

5 Why does a typical electrical cable have a metal wire inside a plastic cover?

- Both parts stop current equally.
- Plastic carries current and metal blocks it.
- Metal carries current; plastic helps insulate it.

> **Answer:** Metal carries current; plastic helps insulate it.
> Metals such as copper are good conductors. The plastic cover is an insulator that helps keep current in the intended path when the cable is intact and used correctly.

6 Two similar pieces of wood are tested. One is dry; the other is wet with ordinary tap water. Which would generally conduct better?

- The wet piece
- The dry piece
- Moisture can never affect wood

> **Answer:** The wet piece
> Moisture containing dissolved ions generally increases wood's conductivity. The exact result also depends on the wood and the test setup.

7 Which sample would generally conduct electricity better: very pure water or water with dissolved table salt?

- Very pure water
- Water with dissolved salt
- Neither can carry any current

> **Answer:** Water with dissolved salt
> Dissolved salt provides charged particles called ions that can carry current. Very pure water conducts poorly.

8 In a working battery-and-bulb test, inserting a metal key makes the bulb light. What does this result show?

- All materials conduct equally well.
- The key creates electricity by itself.
- The key allows enough current to flow to light the bulb.

> **Answer:** The key allows enough current to flow to light the bulb.
> The key helps complete a conducting path. The battery supplies the energy. A bulb that stays dark in another test does not prove that absolutely no current can flow.

9 Air normally insulates well. How can an electrical discharge still pass through it in a storm?

- A very strong electric field can make the air conduct.
- Air is always as good a conductor as copper.
- Air conducts only when every metal object is removed.

> **Answer:** A very strong electric field can make the air conduct.
> A strong enough electric field can break down air's insulating behaviour. The resulting conducting path allows a discharge.

10 You hear thunder outside. Which is the appropriate shelter among these choices?

- A picnic shelter with open sides
- Inside a solid building with closed walls and a roof
- Under a lone tall tree

> **Answer:** Inside a solid building with closed walls and a roof
> Go indoors when you hear thunder. A fully enclosed vehicle with a metal roof and the windows closed is another suitable option. Open shelters and trees do not provide proper protection.

11 The storm seems to have passed. How long should you wait after the last thunder before returning outdoors?

- Until the rain stops
- Just 2 minutes
- At least 30 minutes

> **Answer:** At least 30 minutes
> Wait at least 30 minutes after the last thunder. A storm can still be dangerous after the rain stops.

12 What is the purpose of a properly installed lightning protection system on a building?

- Provide a conducting route to the ground if the building is struck
- Pull every nearby storm toward the roof
- Store the storm inside a metal rod

> **Answer:** Provide a conducting route to the ground if the building is struck
> The system helps carry the discharge to the ground and reduce damage. A lightning rod does not attract lightning or guarantee that a building can never be damaged.

Generated from the sources listed in the quiz definition file.